

Load Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q11-0031

Blackrock 345 kV
August 20, 2012

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0031. The Interconnection Customer proposes a 197.4 MW large load (data center) facility interconnecting at the Blackrock 345 kV substation in the Pennsylvania area. The request was received on April 24, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 27 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$5,906,000.

3. Network Impacts

3.1 Load Deliverability Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
MILESBURG - WOODSIDE 138kV	ORCHARD RD - SHAWVILLE STATION 500kV	120.3	258	5.6
MILESBURG - SCOTT 138kV	ORCHARD ROAD - SHAWVILLE STA 500kV L	108.4	272	5.6

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
MILESBURG - WOODSIDE 138kV	Rebuild line with higher-capacity conductor	\$5,584,000	100% (\$5,584,000)
SCOTT - MILESBURG 138kV	Reconductor line	\$322,000	100% (\$322,000)
	Total allocated to project		\$5,906,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.